AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX EXAMINATION

APRIL/ MAY 2017

Physics Paper II

Time: 2 hours 25 minutes Marks: 40

May Examination 2017

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are ELEVEN questions. Answer ALL questions. Questions 9, 10 & 11 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

   Read each question carefully.
   Use a black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

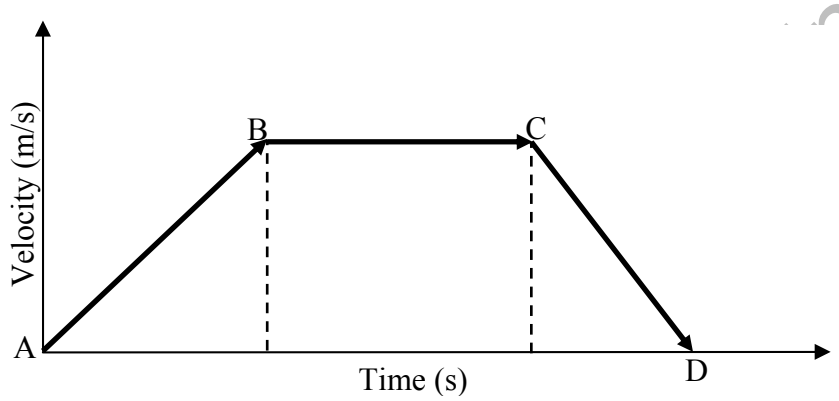
5. You may use a simple calculator if you wish.

Q.1. (Total 3 Marks)

Describe any THREE important contributions of physics in the field of technology.

Q.2. (Total 3 Marks)

Consider the given velocity-time graph for a body.



Interpret the following segments with respect to 'acceleration' from the given graph.

i. A to B (1 Mark)

ii. B to C (1 Mark)

iii. C to D (1 Mark)

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Q.3. (Total 3 Marks)

List any THREE methods of reducing friction.

_____

_____

_____

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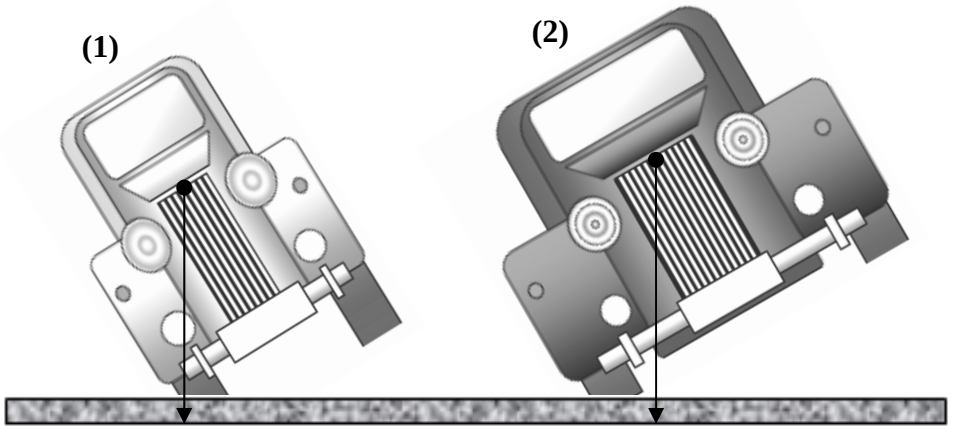
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Q.4. (Total 3 Marks)

Consider the given figure to answer the following questions.



i. Identify the car which is more stable with reference to its centre of gravity (●). (1 Mark)

ii. For the car identified as more stable, write an effect of each of the following on its stability. (2 Marks)

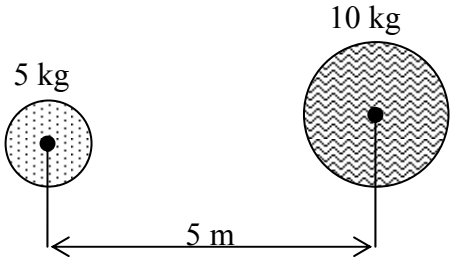
- Position of the centre of gravity
- Area of the base

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Q.5. (Total 3 Marks)

Calculate the force of gravitation between the given balls.

(**Note:** The value of gravitational constant $G = 6.67 \times 10^{-11}$ Nm^2/kg^2.)

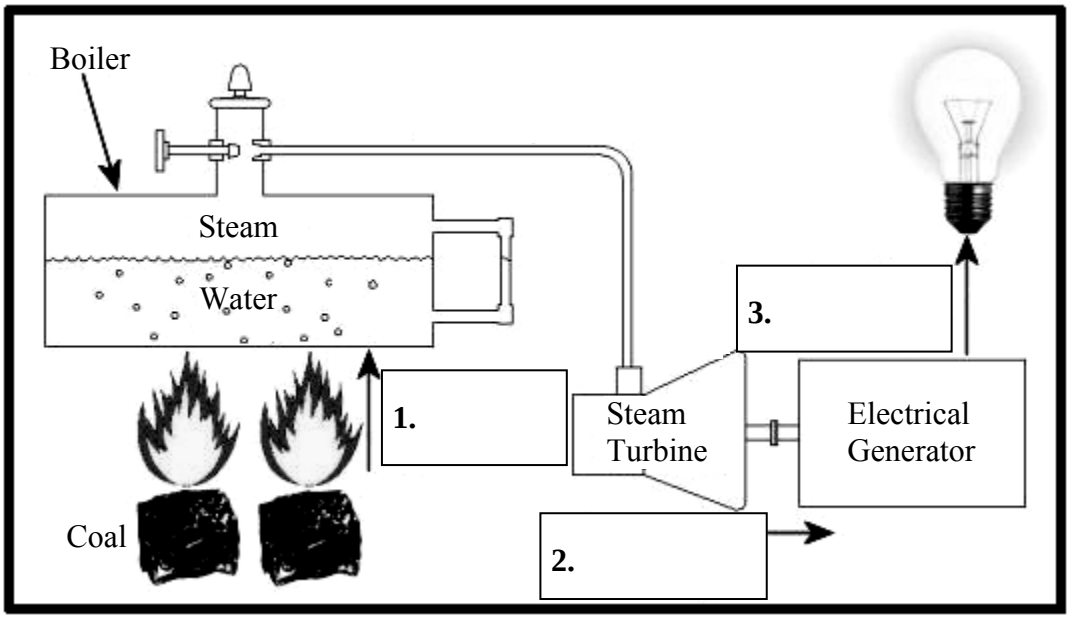


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Q.6. (Total 3 Marks)

Identify the types of energy at positions **1**, **2** and **3** in the given illustration of a thermal plant.



1. _____

2. _____

3. _____

Q.7. (Total 3 Marks)

Calculate the pressure due to fluid at a depth of 10 cm in still water.

(**Note**: The density of water is 1000 kg/m^3 and the value of acceleration due to gravity is 10 m/s^2.)

Q.8. (Total 4 Marks)

A 4 cm thick cast iron sheet has a cross-sectional area of 10,000 cm^2. If one side of the sheet is at 500°C and the other is at 600°C, then how much amount of heat passes through the sheets in each second?

(**Note:** The thermal conductivity 'k' of cast iron is 85 W/m.K)

Q.9. (Total 5 Marks)

EITHER

a. A boat travels toward south with a steady velocity of 30 m/s through a river in which there is a steady flow of water in the direction of the east at a velocity of 15 m/s.

i. Illustrate the given situation using a labelled vector diagram. (1 Mark)

ii. Show the direction of the motion of the boat in the diagram. (1 Mark)

iii. Calculate the resultant velocity of the boat. (3 Marks)

OR

b. An object of mass *m* is given an upward acceleration $\vec{\mathbf{a}}$ by pulling a rope.



i. Label the following points in the given diagram. (2 Marks)

- The direction of weight of the object
- The direction of tension in the rope

ii. If the object is moving upward and the weight is given by $F_w = mg$, then derive a simplified expression for the tension in the rope. (3 Marks)

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Q.10. (Total 5 Marks)

EITHER

a. 'A working system cannot have an efficiency of 100%.'

Justify the given statement with reference to heat expels, heat transfers or Carnot limit.

OR

b. Describe why the atmospheric pressure falls at a height above the sea level. Give your answer with reference to gravity, air compression and temperature of the surrounding.

Q.11. (Total 5 Marks)

EITHER

a. Explain with a daily life example how evaporation causes cooling effect.

OR

b. Discuss any FIVE applications of heat transfer by the method of radiation.

END OF PAPER

Please use this page for rough work

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